

Conversational AI in Games: An Empirical Study of LLM-Powered NPC Dialogue Systems

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Motivation

- **NPC dialogue** is central to *immersion*, *agency*, and *narrative depth*
- **Traditional branching systems** → predictable & repetitive
- **LLMs** enable *dynamic*, *open-ended conversations*
- **Gap:** little empirical research on their *actual impact* in games
- **Need:** compare designs to *guide future dialogue systems*

Research Questions



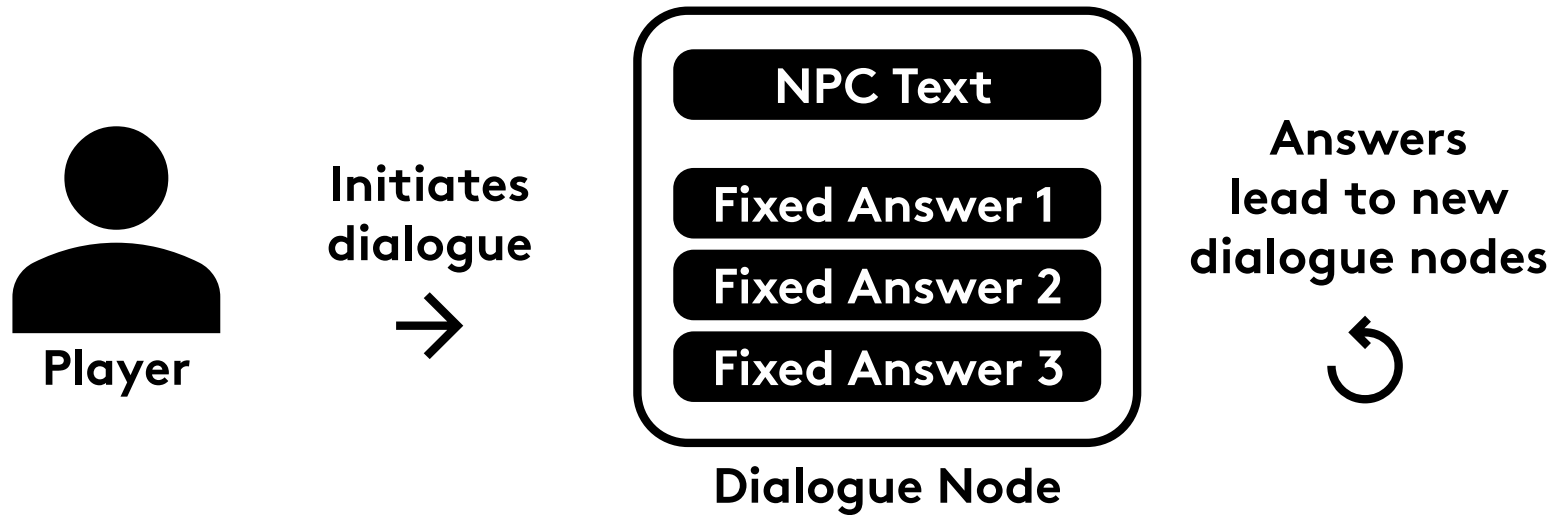
RQ1: How does the integration of different LLM-driven features in NPC dialogues influence player interaction in video games?



RQ2: Did any LLM-driven feature enhance player interaction compared to the Control Version?

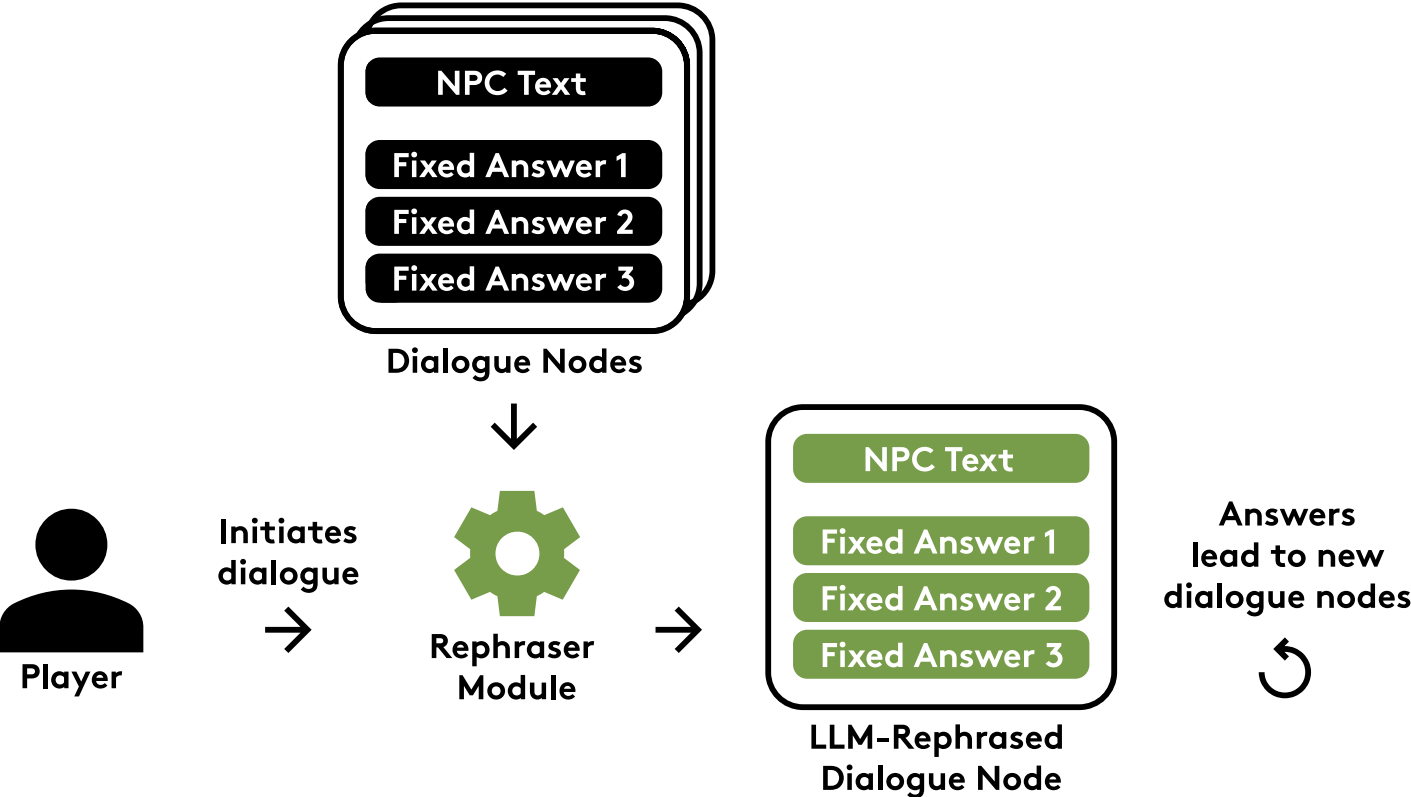
Control Version

→ *Static branching dialogue tree* (traditional RPG style)



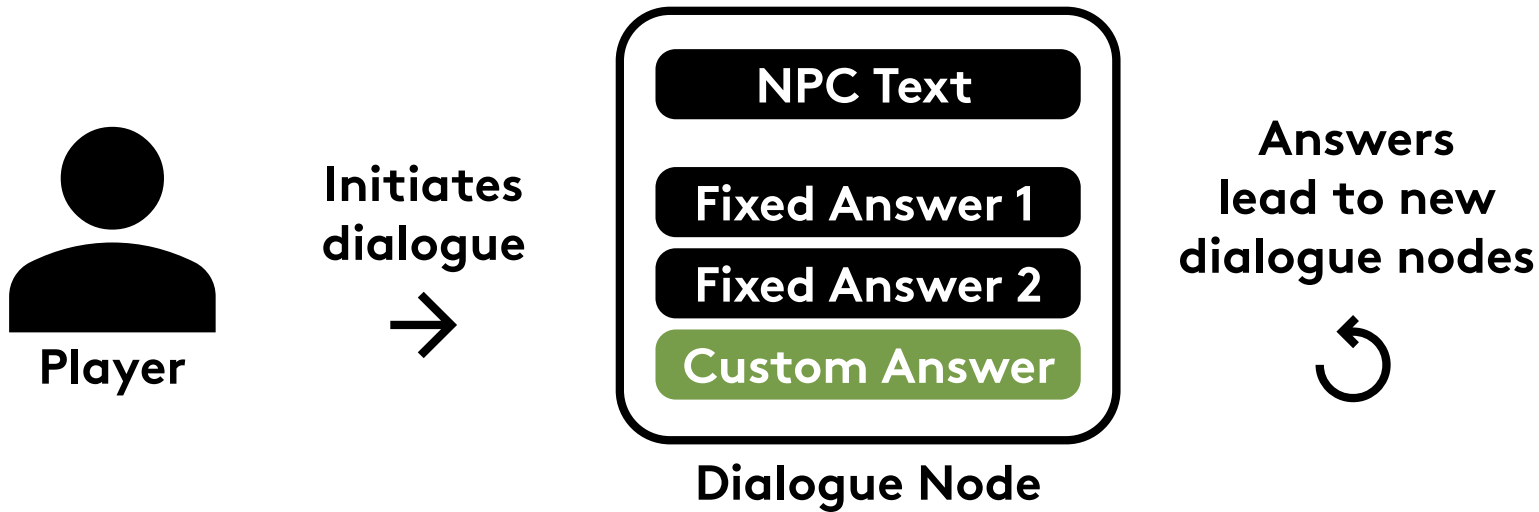
Version A

→ *LLM* rephrases static lines → linguistic variation, but still choice-based



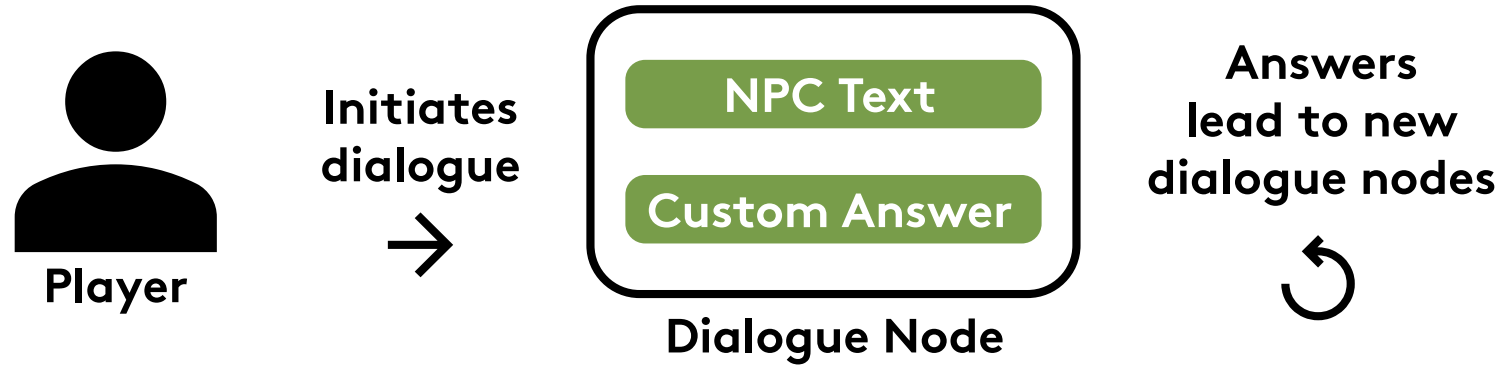
Version B

→ Hybrid system → fixed options + free-text input



Version C

→ *Fully open-ended* → all player input as free text, LLM generates NPC replies



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You
What happened?

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Our town is in dire straits. The food supply was stolen, leaving us all in peril. You should speak to the Innkeeper, she might have some leads.

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Version C



Methodology

- **Prototype:** Custom Unity RPG in medieval setting
- **Consistency:** Same quests, mechanics, and visuals across all versions
- **Participants:** 64 tested (62 analyzed), mean age **25.2**
- **Procedure:** briefing → gameplay (random dialogue system) → questionnaire
- **Goal:** Compare *player behavior + perceptions* across systems



Data Collection

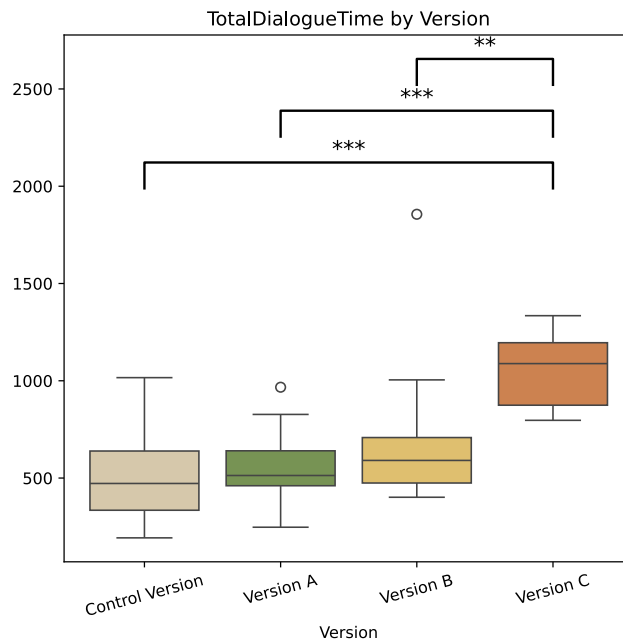
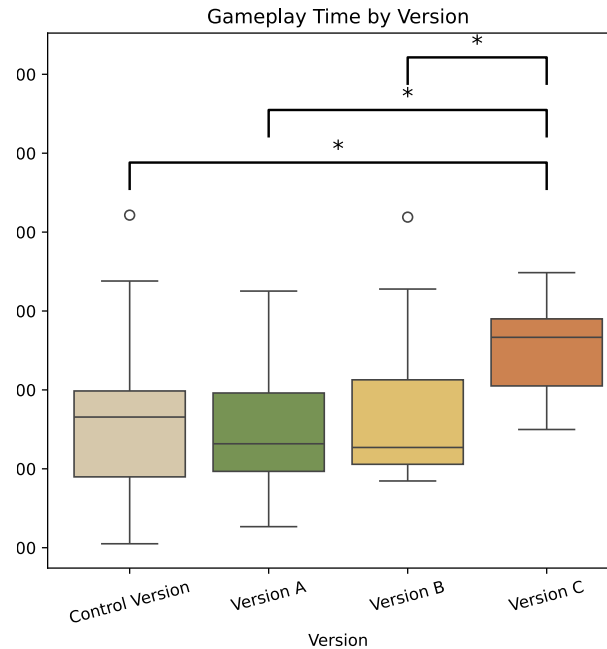
→ Game Logging:

- Dialogue time & number of exchanges
- Movement, area transitions, quest outcomes
- LLM wait/load times, input types (Version B)

→ Survey Responses:

- 24 Likert-scale items + 4 open-ended questions
- Covered *engagement, immersion, clarity, enjoyment*
- Collected *technical data & background* for context

Results: Gameplay & Dialogue Time



→ **Version C** → significantly longer *total gameplay time* ($p = 0.0064$)

→ Extra time came from **dialogues**, not slower gameplay

→ Version C > Control, A, B in *time spent talking*

→ **Casual players**: longer dialogues in Version C, but *same completion time*

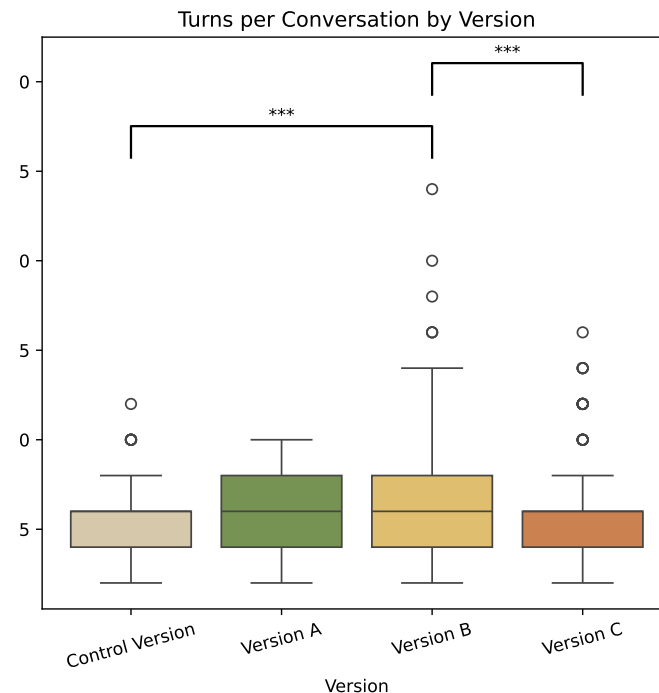
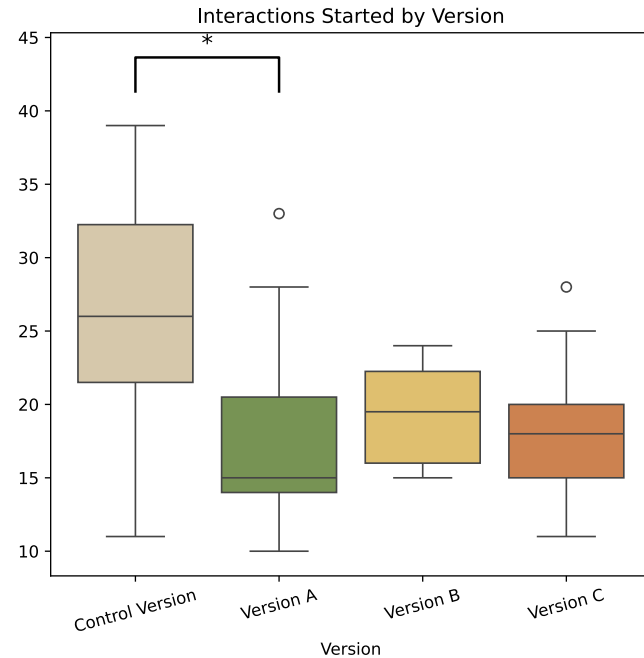
Results: Interaction Quality

→ **Control (CV):** more dialogue initiations than Version A ($p = 0.0105$)

→ **Version A:** fewer initiations, lower engagement

→ **Version B:** longer *turns per interaction* ($p < .0001$)

→ **Version C:** balanced *volume + depth* → most engaging



Survey Question	Control	Version A	Version B	Version C
The game was engaging and kept my interest throughout.	3.94	3.67	4.19	4.20
I found the NPCs interesting to talk to.	3.25	3.00	3.75	4.00
I felt motivated to gather information from NPCs.	3.94	3.80	3.94	3.93
The game's dialogue system made the experience enjoyable.	3.75	3.47	4.12	4.27
I felt in control of my interactions with NPCs.	3.75	3.53	3.12	3.87
The dialogue system allowed for meaningful conversations.	3.31	3.00	3.44	3.60
I felt the NPCs reacted naturally to my dialogue choices.	4.00	3.93	3.44	3.87
NPCs provided key information in an immersive way.	4.00	4.27	3.94	4.00
I felt encouraged to experiment with dialogue.	3.19	3.53	3.94	4.07
There were moments the dialogue felt unnatural or confusing.	1.94	2.00	3.19	2.87
After speaking with NPCs, I knew what to do next.	3.88	4.53	4.25	4.00
Finding and using info from NPCs was engaging.	4.19	3.93	3.94	4.07
NPCs conveyed information consistently.	4.19	4.20	3.75	4.20
I would be interested in playing a longer version of this game.	4.00	3.67	4.50	4.47
I would be interested in games with similar dialogue systems.	3.88	3.60	4.31	4.13
The dialogue system increased my interest in AI-powered NPCs.	3.19	3.07	3.75	3.93

TABLE 5.4: Likert-scale Survey Averages by Game Version

Results: Player Impressions

- **Version C praised:** *“More natural”, “felt like role-playing”*
 - Players valued **freedom** and **expressiveness**
 - Stronger sense of **immersion** compared to fixed options
- **Critiques:** occasional *off-topic answers* or NPCs *pushing dialogue forward*
- Overall: Version C felt **intuitive** and **engaging**, especially for casual players

Discussion

Metric	Control Version	Version A	Version B	Version C
Engagement	+	-	++	+++
Flexibility	+	+	++	+++
Stability	+	+	--	-
Technical Performance	+	-	--	-
Player Agency	+	+	++	+++
Perceived Naturalness	+	+	±	++
Dialogue Control (Dev-side)	+	±	±	--

TABLE 6.1: Comparative summary of each dialogue version across key metrics

- **Control (CV):** stable, predictable → but rigid & less immersive
- **Version A:** minor variation, added load time → limited impact
- **Version B:** hybrid freedom + structure → engaging, but inconsistent
- **Version C:** most engaging & natural → strong immersion, favored casual players
- **Trade-off:** higher player agency vs. stability & developer control

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Write your own answer here...

Send

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The Maersk Mc-Kinney Moller Institute

Future Work & Conclusion

→ Future Work

- **Broader testing** → larger, more diverse player groups and different genres
- **Improve stability** → refine hybrid systems with better prompt design and tuning

→ Conclusion

- **Open-ended dialogue (Version C)** → highest engagement and immersion
- **Casual players** → showed the strongest benefits from natural conversation

Thank you!